

$$\frac{g \, h \, m \, \cos\{\left(\alpha\{\left(t \, \text{right}\right)\} \, \text{right}\)}{2}$$

$$+ 0.5 \text{ m} \left(\frac{h^2}{\sin^2\{\alpha(t)\}} \frac{d^2}{dt^2} \alpha(t) + \frac{h^2}{\cos^2\{\alpha(t)\}} \frac{d^2}{dt^2} \alpha(t) \right)$$

$$\begin{aligned}
& \left(\frac{1}{2} \left(\cos(\alpha) + \cos(\beta) \right) \right)^2 + \frac{1}{2} \left(\cos(\alpha) - \cos(\beta) \right)^2 + \frac{1}{2} \left(\sin(\alpha) + \sin(\beta) \right)^2 + \frac{1}{2} \left(\sin(\alpha) - \sin(\beta) \right)^2 \\
&= \frac{1}{2} \left(\cos^2(\alpha) + 2\cos(\alpha)\cos(\beta) + \cos^2(\beta) \right) + \frac{1}{2} \left(\cos^2(\alpha) - 2\cos(\alpha)\cos(\beta) + \cos^2(\beta) \right) \\
&\quad + \frac{1}{2} \left(\sin^2(\alpha) + 2\sin(\alpha)\sin(\beta) + \sin^2(\beta) \right) + \frac{1}{2} \left(\sin^2(\alpha) - 2\sin(\alpha)\sin(\beta) + \sin^2(\beta) \right) \\
&= \frac{1}{2} \left(2\cos^2(\alpha) + 2\cos^2(\beta) + 2\sin^2(\alpha) + 2\sin^2(\beta) \right) \\
&= \cos^2(\alpha) + \cos^2(\beta) + \sin^2(\alpha) + \sin^2(\beta) \\
&= 2\cos^2(\alpha) + 2\sin^2(\alpha) \\
&= 2\left(\cos^2(\alpha) + \sin^2(\alpha) \right) \\
&= 2 \cdot 1 \\
&= 2
\end{aligned}$$

[illegible]

$$t(\alpha(t\,right)\,right) - 1right)^{2} + 2\sin^{2}(t\,(\alpha(t\,right)\,right)\,right)^{2}) + \frac{\left(\cos\left(\alpha(t\,right)\,right) - 1right)^{2}\right)\left(4\left(\cos\left(\alpha(t\,right)\,right) - 1right)^{2}\right) + 2\sin^{2}(t\,(\alpha(t\,right)\,right)\,right) + \frac{\sin^{2}(t\,(\alpha(t\,right)\,right)\,right)}{2}\left(\cos\left(\alpha(t\,right)\,right) - 1right)^{2}\right) + 2\sin^{2}(t\,(\alpha(t\,right)\,right)\,right)\right)}{\frac{1}{2} - \frac{\cos\left(\alpha(t\,right)\,right) - 1right)^{2}}{2}\left(\cos\left(\alpha(t\,right)\,right) - 1right)^{2}\right) + 2\sin^{2}(t\,(\alpha(t\,right)\,right)\,right)} + \frac{\sin^{2}(t\,(\alpha(t\,right)\,right)\,right)}{2}\left(\cos\left(\alpha(t\,right)\,right) - 1right)^{2}\right) + 2\sin^{2}(t\,(\alpha(t\,right)\,right)\,right)\,right)\left(-0.0709148443026524\,h - w + 5right)\left(-\frac{\left(\cos\left(\alpha(t\,right)\,right) - 1right)\sin\left(\alpha(t\,right)\,right) - 4\sin\left(\alpha(t\,right)\,right)\cos\left(\alpha(t\,right)\,right)\right)\left(\cos\left(\alpha(t\,right)\,right) - 1right)^{2}\right)\left(4\left(\cos\left(\alpha(t\,right)\,right) - 1right)^{2}\right) + 2\sin^{2}(t\,(\alpha(t\,right)\,right)\,right)^{2}\right) - \frac{\left(2\left(\cos\left(\alpha(t\,right)\,right) - 1right)\sin\left(\alpha(t\,right)\,right) - 4\sin\left(\alpha(t\,right)\,right)\cos\left(\alpha(t\,right)\,right)\right)\sin^{2}(t\,(\alpha(t\,right)\,right)\,right)}{2\left(\cos\left(\alpha(t\,right)\,right) - 1right)^{2}\right) + 2\sin^{2}(t\,(\alpha(t\,right)\,right)\,right))wig$$

[illegible]

$$\begin{aligned}
& -1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)+\frac{\sin ^{2}\left(\alpha \left(t\right)\right)}{2}\left(\cos \left(\alpha \left(t\right)\right)-1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)\right)^{\frac{3}{2}}-\frac{1}{2}\sqrt{2}h\left(\cos \left(\alpha \left(t\right)\right)-1\right)\sin \left(\alpha \left(t\right)\right)-4\sin \left(\alpha \left(t\right)\right)\cos \left(\alpha \left(t\right)\right)\right)\sin ^{3}\left(\alpha \left(t\right)\right)\left(\sqrt{\cos \left(\alpha \left(t\right)\right)-1}\right)^{2}\left(2\left(\cos \left(\alpha \left(t\right)\right)-1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)+\frac{\sin ^{2}\left(\alpha \left(t\right)\right)}{2}\left(\cos \left(\alpha \left(t\right)\right)-1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)\right)\left(\cos \left(\alpha \left(t\right)\right)-1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)-\frac{5}{2}\sqrt{2}h\sin ^{2}\left(\alpha \left(t\right)\right)\cos \left(\alpha \left(t\right)\right)\left(\sqrt{\cos \left(\alpha \left(t\right)\right)-1}\right)^{2}\left(2\left(\cos \left(\alpha \left(t\right)\right)-1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)+\frac{\sin ^{2}\left(\alpha \left(t\right)\right)}{2}\left(\cos \left(\alpha \left(t\right)\right)-1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)\right)\left(\cos \left(\alpha \left(t\right)\right)-1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)-\frac{1}{2}\sqrt{2}h\left(\frac{\cos \left(\alpha \left(t\right)\right)-1}{2}\left(2\left(\cos \left(\alpha \left(t\right)\right)-1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)+\frac{\sin ^{2}\left(\alpha \left(t\right)\right)}{2}\left(\cos \left(\alpha \left(t\right)\right)-1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)\right)\left(\cos \left(\alpha \left(t\right)\right)-1\right)^{2}+2\sin ^{2}\left(\alpha \left(t\right)\right)\right)
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{2} \left(\cos(\alpha) \cos(\beta) + \sin(\alpha) \sin(\beta) \right) \left(\cos(\alpha) \cos(\beta) + \sin(\alpha) \sin(\beta) \right) \\
&= \frac{1}{2} \left(\cos^2(\alpha) \cos^2(\beta) + \cos^2(\alpha) \sin^2(\beta) + \sin^2(\alpha) \cos^2(\beta) + \sin^2(\alpha) \sin^2(\beta) \right) \\
&= \frac{1}{2} \left(\cos^2(\alpha) (\cos^2(\beta) + \sin^2(\beta)) + \sin^2(\alpha) (\cos^2(\beta) + \sin^2(\beta)) \right) \\
&= \frac{1}{2} (\cos^2(\alpha) + \sin^2(\alpha)) (\cos^2(\beta) + \sin^2(\beta)) \\
&= \frac{1}{2} (1) (1) \\
&= \frac{1}{2}
\end{aligned}$$

$$\begin{aligned} & \left(\cos(\alpha(t)) - 1\right)^2 + 2\sin(\alpha(t))\cos(\alpha(t)) + \frac{\sin^2(\alpha(t))}{\cos(\alpha(t)) - 1} \left(\cos(\alpha(t)) - 1\right)^2 + 2\sin(\alpha(t))\cos(\alpha(t)) \\ & \left(\frac{2\cos(\alpha(t)) - 1}{\sin(\alpha(t))} \sin(\alpha(t)) - 4\sin(\alpha(t))\cos(\alpha(t)) \sin(\alpha(t)) \cos(\alpha(t)) \sin(\alpha(t)) \right. \\ & \left. - \frac{2\cos(\alpha(t)) - 1}{\sin(\alpha(t))} \sin(\alpha(t)) \cos(\alpha(t)) \sin(\alpha(t)) \right) \left(\cos(\alpha(t)) - 1\right)^2 + 2\sin(\alpha(t))\cos(\alpha(t)) \\ & \left(\frac{2\cos(\alpha(t)) - 1}{\sin(\alpha(t))} \sin(\alpha(t)) - 4\sin(\alpha(t))\cos(\alpha(t)) \sin(\alpha(t)) \cos(\alpha(t)) \sin(\alpha(t)) \right. \\ & \left. - \frac{2\cos(\alpha(t)) - 1}{\sin(\alpha(t))} \sin(\alpha(t)) \cos(\alpha(t)) \sin(\alpha(t)) \right) \left(\cos(\alpha(t)) - 1\right)^2 + 2\sin(\alpha(t))\cos(\alpha(t)) \end{aligned}$$

$$\begin{aligned}
& \text{right})\}\text{right})^{\wedge}\{2\} + \text{frac}\{\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}^{\wedge}\{2\}\}\{4\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}^{\wedge}\{2\} + 2\sin^{\wedge}\{2\}\{\text{left}\{\alpha\text{right}\}\text{right}\}\text{right}\} + \text{frac}\{\sin^{\wedge}\{2\}\{\text{left}\{\alpha\text{right}\}\text{right}\}\}\{2\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}^{\wedge}\{2\} + 2\sin^{\wedge}\{2\}\{\text{left}\{\alpha\text{right}\}\text{right}\}\text{right}\}\} + \text{frac}\{\text{sqrt}\{2\}\text{w}\text{left}\{1 - \cos\{\text{left}\{\alpha\text{right}\}\text{right}\}\}\text{right}\}\text{left}\{-\text{frac}\{\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}^{\wedge}\{2\}\}\{2\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}^{\wedge}\{2\} + 2\sin^{\wedge}\{2\}\{\text{left}\{\alpha\text{right}\}\text{right}\}\text{right}\} + \text{frac}\{\sin^{\wedge}\{2\}\{\text{left}\{\alpha\text{right}\}\text{right}\}\}\{\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}^{\wedge}\{2\} + 2\sin^{\wedge}\{2\}\{\text{left}\{\alpha\text{right}\}\text{right}\}\text{right}\}\text{right}\}\text{left}\{\text{frac}\{2\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}\sin\{\text{left}\{\alpha\text{right}\}\text{right}\} - 4\sin\{\text{left}\{\alpha\text{right}\}\text{right}\}\cos\{\text{left}\{\alpha\text{right}\}\text{right}\}\text{right}\}\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}^{\wedge}\{2\}\}\{\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}^{\wedge}\{2\} + 2\sin^{\wedge}\{2\}\{\text{left}\{\alpha\text{right}\}\text{right}\}\text{right}\}\} - \text{frac}\{2\text{left}\{2\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}\sin\{\text{left}\{\alpha\text{right}\}\text{right}\} - 4\sin\{\text{left}\{\alpha\text{right}\}\text{right}\}\cos\{\text{left}\{\alpha\text{right}\}\text{right}\}\text{right}\}\sin^{\wedge}\{2\}\{\text{left}\{\alpha\text{right}\}\text{right}\}\}\{\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}^{\wedge}\{2\} + 2\sin^{\wedge}\{2\}\{\text{left}\{\alpha\text{right}\}\text{right}\}\text{right}\}\} - \text{frac}\{2\text{left}\{\text{cos}\{\text{left}\{\alpha\text{right}\}\text{right}\} - 1\text{right}\}^{\wedge}\{2\} + 2\sin^{\wedge}\{2\}\{\text{left}\{\alpha\text{right}\}\text{right}\}\text{right}\}\} - 1
\end{aligned}$$

$$\begin{aligned} & \left(\text{left}(\cos(\text{left}(\alpha(\text{left}(t \text{ right})) \text{ right})) - 1 \text{ right})^{\wedge} \{2\} + 2 \sin^{\wedge} \{2\}(\text{left}(\alpha(\text{left}(t \text{ right})) \text{ right})) \right) + \text{frac}(\sin^{\wedge} \{2\}(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) \{2 \text{ left}(\cos(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) - 1 \text{ right})^{\wedge} \{2\} + 2 \sin^{\wedge} \{2\}(\text{left}(\alpha(\text{left}(t \text{ right})) \text{ right})) \right)\} + \text{frac}(\sqrt{2} \text{ h left}(1 - \cos(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) \text{ left}(2 \cos(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) - 1 \text{ right}) \sin(\text{left}(\alpha(\text{left}(t \text{ right})) \text{ right})) - 4 \sin(\text{left}(\alpha(\text{left}(t \text{ right})) \text{ right}) \cos(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) \cos^{\wedge} \{2\}(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) \{ \text{left}(\cos(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) - 1 \text{ right})^{\wedge} \{2\} + 2 \sin^{\wedge} \{2\}(\text{left}(\alpha(\text{left}(t \text{ right})) \text{ right})) \right)\} \sqrt{2} \text{ left}(\frac{\cos(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) - 1 \text{ right})^{\wedge} \{2\}}{2 \text{ left}(\cos(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) - 1 \text{ right})^{\wedge} \{2\} + 2 \sin^{\wedge} \{2\}(\text{left}(\alpha(\text{left}(t \text{ right})) \text{ right})) \right)\} + \text{frac}(\sin^{\wedge} \{2\}(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) \{ \cos(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) - 1 \text{ right})^{\wedge} \{2\} + 2 \sin^{\wedge} \{2\}(\text{left}(\alpha(\text{left}(t \text{ right})) \text{ right})) \right)\} \{4 \text{ left}(\cos(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) - 1 \text{ right})^{\wedge} \{2\} + 2 \sin^{\wedge} \{2\}(\text{left}(\alpha(\text{left}(t \text{ right})) \text{ right})) \right)\} + \text{frac}(\sin^{\wedge} \{2\}(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) \{2 \text{ left}(\cos(\text{left}(\alpha(\alpha(\text{left}(t \text{ right})) \text{ right})) - 1 \text{ right}) \sin(\text{left}(\alpha(\text{left}(t \text{ right})) \text{ right})) - 4 \sin(\text{le} \end{aligned}$$

